

Lab 1: Requirements Table

Scenario: Multi-Cloud Asset Cost Optimization Portal

Target Stakeholders / Actors: FinOps Analyst, Cloud Administrator (Cloud Billing System as external actor supplying data).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest cloud billing CSV exports from AWS, Azure, and GCP, categorize costs by service and project tag, and highlight compute instances with < 5% CPU utilization over 14 days.	High	Pass: Idle cloud asset flagged with monthly savings estimate. Fail: Active instance misclassified as idle.	Enables analysts to spot wasted compute spend without manual log review.
FR-002	Functional	The system shall identify storage volumes that have been unattached to any compute instance for more than 7 consecutive days and flag them for review or deletion.	High	Pass: Volume unattached for >7 days appears on the Idle Assets list with the days-unattached count. Fail: An attached volume is listed, or an eligible volume is omitted.	Unattached storage is a recurring, easily-automated source of avoidable cloud spend.
FR-003	Functional	The system shall dispatch an automated budget breach alert (email and/or Slack) to the responsible FinOps Analyst within 5 minutes of a project's monthly spend exceeding its configured threshold.	High	Pass: Alert is received by the analyst within 5 minutes of the breach being detected, including project name and overage amount. Fail: No alert sent, or alert sent >5 minutes after breach, or missing overage detail.	Timely alerts let teams act before overspend compounds across a billing cycle.
FR-004	Functional	The system shall allow a FinOps Analyst to create, view, and update a monthly budget threshold for each cloud project or cost-center tag.	Medium	Pass: A threshold saved by the analyst is applied to that project's breach detection on the next ingestion cycle. Fail: Saved threshold is not applied, or is applied to the wrong project.	Thresholds vary by team and must be analyst-configurable rather than hard-coded.
FR-005	Functional	The system shall generate a monthly cost-optimization report, exportable as PDF or CSV, summarizing idle/unattached assets found and their estimated monthly savings.	Medium	Pass: Exported report lists every asset flagged idle that month with a savings estimate, and totals reconcile with the dashboard. Fail: Report omits a flagged asset or totals do not reconcile.	Gives stakeholders an auditable, shareable record for cost-review meetings.
NFR-001	Nonfunctional	The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Dashboard update exceeds 2 minutes for a multi-GB file under the same test.	Analysts rely on near-real-time dashboards; slow refreshes delay cost decisions.
NFR-002	Nonfunctional (Security)	The system shall enforce role-based access control (RBAC) so only authorized Cloud Administrators can upload/ingest billing files and only authorized FinOps Analysts can configure budget thresholds and view cost data.	High	Pass: A user without the required role receives a 403/Access Denied response and the attempt is logged. Fail: An unauthorized user can view billing data or change a threshold.	Billing data is sensitive; unrestricted access risks data leakage and accidental misconfiguration.